WARL

Weighted Association Rule Learning for HCV-Human Protein Interaction Prediction

Project completion report

This report documents the implemented pipeline, the real-data run, generated results, validation, and remaining scientific caveats.

Executive summary

This project implements a readable Python pipeline inspired by the paper *A Weighted Association Rule Mining Method for Predicting HCV-Human Protein Interactions*. It starts from a curated HCV-human interaction network, finds repeated interaction patterns with BiMax biclustering, converts those patterns into association rules, and applies Gene Ontology molecular-function information to rank candidate interactions.

The completed run processed the repository dataset and produced 1,900 candidate interactions. All 1,900 were absent from the input interaction list, so they are novel relative to this dataset. The output is a research prototype and should be treated as a ranked hypothesis list, not as experimentally confirmed biology.

Metric	Value
Known input interactions	591
Human proteins	527
HCV proteins	10
Generated predictions	1,900
Novel relative to input	1,900
Automated tests	15 passed

Methodology

1. Data preparation: the VirHostNet-derived CSV is loaded, duplicate edges are removed, and a binary human-by-viral incidence matrix (HV) plus its transpose (VH) are constructed.

2. Biclustering: the custom BiMax implementation extracts closed all-ones submatrices, representing repeated interaction patterns.

3. Rule mining: closed patterns are expanded into small itemsets, then converted into one-consequent association rules with support and confidence.

4. Biological weighting: GO molecular-function annotations are retrieved from UniProt and cached locally. The paper-inspired score combines rule confidence with GO-term co-occurrence probabilities and a missing-term penalty.

5. Prediction assembly: HV and VH results are merged, duplicate human-viral pairs are removed, and the final table is sorted by score.

Implementation and reproducibility

The implementation is organized as small Python modules under `src/warl/`. The local `.venv` contains the pinned project dependencies. The test suite covers data loading, matrix construction, BiMax behavior, rule generation, itemset expansion, and GO scoring.

Component	Purpose
data_loading.py	CSV loading and HV/VH matrix construction
bimax.py	Closed-itemset biclustering
itemset_expansion.py	Candidate subset generation
rule_mining.py	Support, confidence, and rules
go_weighting.py	GO-aware weighted scoring
pipeline.py	End-to-end orchestration
predictions.csv	Generated ranked results

To reproduce the result, activate the environment and run: `python scripts\run_pipeline.py data\raw\virhostnet_hcv_human_interactions.csv > data\predictions.csv`. UniProt responses are cached in `data/cache/go_annotations.json`.

Results

The output contains 1,900 deduplicated pairs: 1,808 from HV analysis and 92 from VH analysis. The highest-ranked predictions have scores between 0.833 and 0.667 within the top ten results.

Human	HCV	Score	Conf.	Dir.
Q9BYW2	NS4A	0.8333	0.833	VH
P13378	E1	0.8333	0.833	VH
Q9BYW2	NS3	0.6667	0.667	VH
Q9BV73	p7	0.6667	0.667	VH
P13378	NS3	0.6667	0.667	VH
Q96S52	NS2	0.6667	0.667	VH
P04275	p7	0.6667	0.667	VH
O94813	CORE	0.6667	0.667	VH
Q9H8S9	NS3	0.6667	0.667	VH
Q12904	CORE	0.6667	0.667	VH

Interpretation and limitations

The predictions are computational hypotheses. A pair being absent from the input network means only that it was not present in this dataset; it does not prove that the interaction is biologically new.

The project uses a VirHostNet-derived dataset rather than the original HCVpro snapshot used in the 2018 paper. Differences in curation, protein identifiers, and GO annotations make direct numerical comparison imperfect.

The current implementation applies a deterministic candidate cap during expansion of very large VH biclusters. This prevents combinatorial explosion for 527 human-protein columns and should be documented when comparing future runs.

The paper's external enrichment workflow and laboratory validation are outside this codebase. The output should therefore be used to select candidates for further biological review or experimental testing.

Conclusion

The repository now contains a working, test-covered, reproducible implementation of the proposed HCV-human interaction mining workflow. It produces a ranked prediction file and caches external annotations so future runs are faster and auditable. The next research step is not more software setup; it is scientific review of the ranked candidates and a careful comparison with the source paper.

Generated from the repository state on 22 October 2026.